Appendix

A. Dynamic Uplink and Downlink Environment Experiments

Dynamic SNR Fluctuation Under Identical Uplink and Downlink

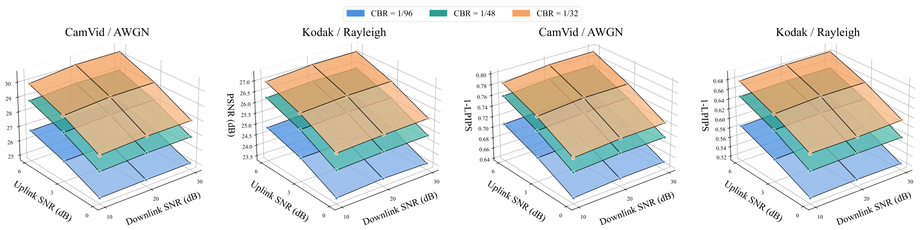

Fig.11. HAMJSCC performance under fluctuating SNR in uplink and downlink.

Fig. 11 plots reconstruction performance across different uplink/downlink SNR. Uplink CBR is 1/96, 1/48, 1/32 and downlink CBR is fixed at 1/24. PSNR and 1-LPIPS rise with higher SNR. On CamVid-AWGN, PSNR increases from 25.03 dB to 30.35 dB, verifying that higher semantic capacity improves reconstruction quality. Uplink SNR has a greater influence on performance than downlink SNR. System quality depends primarily on uplink semantic feature transmission, and downlink latent variables show good compression robustness. Despite performance degradation over Rayleigh fading channels for Kodak, the proposed method works stably.

Impact of Different Uplink and Downlink Environments

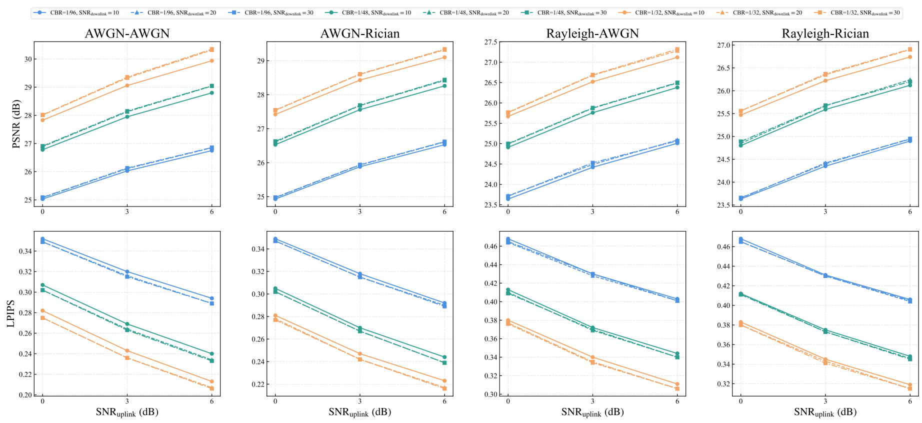

Fig.12. HAMJSCC performance under different uplink and downlink.

As shown in Fig. 12, uplink quality exerts a far more significant impact on system performance, while variations in the downlink lead to relatively limited performance fluctuations. For all four channel combinations, PSNR rises noticeably and LPIPS decreases continuously as the uplink SNR increases from 0 dB to 6 dB. Taking the AWGN-AWGN scenario with CBR = 1/32 as an example, the PSNR increases from 28.01 dB to 30.32 dB, achieving a gain of 2.31 dB. Meanwhile, LPIPS drops from 0.275 to 0.207. In contrast, raising the downlink SNR from 10 dB to 30 dB only brings marginal performance improvement. This result demonstrates that the system possesses strong robustness against downlink channel noise, and the overall reconstruction performance is mainly dominated by the transmission quality of the uplink.

B. Complete Numerical Results for Main Figures

Complete Numerical Results for Fig. 4

CamVid AWGN CBR=1/32

Scheme	PSNR	MS-SSIM
DeepJSCC	[26.23, 27.46, 28.53, 29.47, 30.05, 30.49]	[0.851, 0.891, 0.918, 0.938, 0.951, 0.960]
ADJSCC	[26.54, 27.75, 28.99, 30.08, 30.90, 31.52]	[0.860, 0.900, 0.928, 0.946, 0.960, 0.969]
DeepJSCC-l++	[27.73, 29.15, 30.33, 31.24, 31.87, 32.28]	[0.898, 0.929, 0.949, 0.962, 0.971, 0.976]
SwinJSCC	[27.99, 29.39, 30.49, 31.32, 31.89, 32.24]	[0.903, 0.933, 0.953, 0.966, 0.973, 0.978]
HAMJSCC-Base	[28.28, 29.78, 30.99, 31.88, 32.49, 32.88]	[0.907, 0.938, 0.957, 0.970, 0.977, 0.981]

Kodak AWGN CBR=1/24

Scheme	PSNR	MS-SSIM
DeepJSCC	[25.61, 26.88, 27.85, 28.86, 29.55, 30.10]	[0.830, 0.880, 0.913, 0.937, 0.952, 0.962]
ADJSCC	[25.83, 27.08, 28.00, 28.92, 29.72, 30.44]	[0.836, 0.889, 0.922, 0.936, 0.956, 0.965]
DeepJSCC-l++	[27.00, 28.20, 29.21, 30.01, 30.57, 30.94]	[0.874, 0.910, 0.935, 0.951, 0.961, 0.968]
SwinJSCC	[27.14, 28.49, 29.54, 30.32, 30.85, 31.17]	[0.876, 0.915, 0.940, 0.956, 0.966, 0.971]
HAMJSCC-Base	[27.48, 28.87, 29.98, 30.83, 31.41, 31.77]	[0.881, 0.920, 0.945, 0.960, 0.969, 0.974]

CamVid Rayleigh CBR=1/32

Scheme	PSNR	MS-SSIM
DeepJSCC	[24.82, 25.67, 26.21, 26.67, 26.99, 27.22]	[0.816, 0.855, 0.881, 0.897, 0.907, 0.915]
ADJSCC	[25.27, 26.44, 27.09, 27.48, 27.69, 27.82]	[0.818, 0.876, 0.904, 0.919, 0.929, 0.934]
DeepJSCC-l++	[26.32, 27.34, 28.10, 28.61, 28.93, 29.13]	[0.868, 0.900, 0.921, 0.935, 0.944, 0.949]
SwinJSCC	[26.64, 27.67, 28.37, 28.83, 29.11, 29.27]	[0.880, 0.911, 0.930, 0.942, 0.949, 0.953]
HAMJSCC-Base	[26.79, 27.91, 28.71, 29.25, 29.60, 29.80]	[0.883, 0.915, 0.935, 0.948, 0.955, 0.960]

Kodak Rayleigh CBR=1/24

Scheme	PSNR	MS-SSIM
DeepJSCC	[24.92, 25.88, 26.55, 26.99, 27.42, 27.67]	[0.793, 0.847, 0.878, 0.899, 0.911, 0.920]
ADJSCC	[25.31, 26.26, 26.93, 27.45, 27.80, 27.96]	[0.805, 0.858, 0.890, 0.907, 0.919, 0.925]
DeepJSCC-l++	[26.03, 26.93, 27.62, 28.12, 28.44, 28.62]	[0.840, 0.877, 0.902, 0.918, 0.928, 0.934]
SwinJSCC	[26.33, 27.30, 27.98, 28.41, 28.67, 28.81]	[0.853, 0.891, 0.914, 0.928, 0.936, 0.940]
HAMJSCC-Base	[26.35, 27.42, 28.17, 28.64, 28.95, 29.12]	[0.850, 0.891, 0.916, 0.930, 0.939, 0.943]

Complete Numerical Results for Fig. 5

CamVid AWGN SNR=3

Scheme	PSNR	MS-SSIM
DeepJSCC	[24.49, 25.82, 27.46, 28.22, 28.82, 29.28]	[0.810, 0.855, 0.891, 0.905, 0.915, 0.921]
ADJSCC	[24.67, 26.22, 27.75, 28.80, 29.49, 29.94]	[0.810, 0.864, 0.900, 0.915, 0.923, 0.929]
DeepJSCC-l++	[25.42, 27.73, 29.15, 30.08, 30.67, 31.04]	[0.855, 0.909, 0.929, 0.939, 0.946, 0.950]
SwinJSCC	[25.80, 28.04, 29.39, 30.40, 31.11, 31.47]	[0.876, 0.918, 0.933, 0.942, 0.948, 0.952]
HAMJSCC-Base	[26.23, 28.38, 29.78, 30.78, 31.51, 31.89]	[0.881, 0.921, 0.938, 0.947, 0.953, 0.958]

Kodak AWGN SNR=3

Scheme	PSNR	MS-SSIM
DeepJSCC	[23.95, 25.09, 26.13, 26.88, 27.33, 27.69]	[0.760, 0.820, 0.860, 0.880, 0.892, 0.902]
ADJSCC	[24.32, 25.47, 26.33, 27.08, 27.44, 27.96]	[0.777, 0.838, 0.867, 0.889, 0.901, 0.910]
DeepJSCC-l++	[24.59, 26.41, 27.47, 28.20, 28.67, 28.95]	[0.792, 0.864, 0.894, 0.910, 0.920, 0.927]
SwinJSCC	[24.82, 26.65, 27.63, 28.49, 29.05, 29.37]	[0.820, 0.878, 0.901, 0.915, 0.925, 0.931]
HAMJSCC-Base	[25.40, 27.09, 28.11, 28.87, 29.43, 29.75]	[0.834, 0.885, 0.907, 0.920, 0.929, 0.936]

CamVid Rayleigh SNR=9

Scheme	PSNR	MS-SSIM
DeepJSCC	[24.27, 25.78, 26.67, 27.48, 28.12, 28.66]	[0.825, 0.874, 0.897, 0.912, 0.923, 0.931]
ADJSCC	[24.79, 26.33, 27.48, 28.31, 28.88, 29.27]	[0.852, 0.898, 0.919, 0.930, 0.938, 0.942]
DeepJSCC-l++	[25.45, 27.42, 28.61, 29.39, 29.82, 29.93]	[0.875, 0.919, 0.935, 0.943, 0.947, 0.949]
SwinJSCC	[25.59, 27.64, 28.83, 29.74, 30.37, 30.46]	[0.883, 0.925, 0.942, 0.952, 0.958, 0.960]
HAMJSCC-Base	[26.01, 27.99, 29.25, 30.12, 30.72, 31.05]	[0.892, 0.931, 0.948, 0.956, 0.961, 0.965]

Kodak Rayleigh SNR=9

Scheme	PSNR	MS-SSIM
DeepJSCC	[24.31, 25.57, 26.40, 26.99, 27.61, 28.09]	[0.796, 0.853, 0.883, 0.899, 0.910, 0.919]
ADJSCC	[24.51, 25.75, 26.60, 27.45, 27.94, 28.29]	[0.806, 0.861, 0.886, 0.907, 0.920, 0.924]
DeepJSCC-l++	[24.98, 26.55, 27.47, 28.12, 28.48, 28.57]	[0.828, 0.883, 0.906, 0.918, 0.924, 0.927]
SwinJSCC	[25.05, 26.71, 27.63, 28.41, 28.94, 29.03]	[0.837, 0.890, 0.913, 0.928, 0.937, 0.939]
HAMJSCC-Base	[25.43, 27.03, 27.97, 28.64, 29.15, 29.40]	[0.848, 0.897, 0.918, 0.930, 0.938, 0.943]

Complete Numerical Results for Fig. 6

DeepJSCC-l++ CamVid AWGN

CBR	SNR	PSNR	MS-SSIM
1/96	[0, 3, 6, 9, 12, 15]	[24.21, 25.42, 26.28, 26.88, 27.28, 27.51]	[0.793, 0.855, 0.892, 0.915, 0.930, 0.940]
1/48	[0, 3, 6, 9, 12, 15]	[26.43, 27.73, 28.76, 29.51, 30.02, 30.34]	[0.870, 0.909, 0.933, 0.950, 0.960, 0.967]
1/32	[0, 3, 6, 9, 12, 15]	[27.73, 29.15, 30.33, 31.24, 31.87, 32.28]	[0.898, 0.929, 0.949, 0.962, 0.971, 0.976]
1/24	[0, 3, 6, 9, 12, 15]	[28.59, 30.08, 31.37, 32.39, 33.14, 33.64]	[0.913, 0.939, 0.956, 0.968, 0.975, 0.980]
5/96	[0, 3, 6, 9, 12, 15]	[29.17, 30.67, 31.98, 33.03, 33.80, 34.32]	[0.923, 0.946, 0.961, 0.971, 0.978, 0.982]
1/16	[0, 3, 6, 9, 12, 15]	[29.58, 31.04, 32.28, 33.26, 33.96, 34.41]	[0.930, 0.950, 0.964, 0.973, 0.979, 0.983]

DeepJSCC-l++ Kodak Rayleigh

CBR	SNR	PSNR	MS-SSIM
1/96	[0, 3, 6, 9, 12, 15]	[22.74, 23.82, 24.51, 24.98, 25.26, 25.43]	[0.680, 0.750, 0.797, 0.828, 0.846, 0.857]
1/48	[0, 3, 6, 9, 12, 15]	[24.48, 25.41, 26.07, 26.55, 26.83, 27.01]	[0.776, 0.826, 0.860, 0.883, 0.897, 0.905]
1/32	[0, 3, 6, 9, 12, 15]	[25.42, 26.34, 26.99, 27.47, 27.78, 27.97]	[0.816, 0.859, 0.887, 0.906, 0.917, 0.924]
1/24	[0, 3, 6, 9, 12, 15]	[26.03, 26.93, 27.62, 28.12, 28.44, 28.62]	[0.840, 0.877, 0.902, 0.918, 0.928, 0.934]

5/96	[0, 3, 6, 9, 12, 15]	[26.40, 27.30, 27.99, 28.48, 28.80, 28.99]	[0.854, 0.887, 0.910, 0.924, 0.933, 0.938]
1/16	[0, 3, 6, 9, 12, 15]	[26.64, 27.49, 28.13, 28.57, 28.85, 29.02]	[0.863, 0.894, 0.914, 0.927, 0.935, 0.939]

SwinJSCC CamVid AWGN

CBR	SNR	PSNR	MS-SSIM
1/96	[0, 3, 6, 9, 12, 15]	[24.78, 25.80, 26.49, 26.94, 27.21, 27.37]	[0.828, 0.876, 0.904, 0.921, 0.932, 0.938]
1/48	[0, 3, 6, 9, 12, 15]	[26.80, 28.04, 28.98, 29.64, 30.09, 30.35]	[0.882, 0.918, 0.940, 0.954, 0.963, 0.969]
1/32	[0, 3, 6, 9, 12, 15]	[27.99, 29.39, 30.49, 31.32, 31.89, 32.24]	[0.903, 0.933, 0.953, 0.966, 0.973, 0.978]
1/24	[0, 3, 6, 9, 12, 15]	[28.86, 30.40, 31.70, 32.71, 33.44, 33.90]	[0.915, 0.942, 0.960, 0.972, 0.979, 0.983]
5/96	[0, 3, 6, 9, 12, 15]	[29.48, 31.11, 32.49, 33.60, 34.42, 34.95]	[0.923, 0.948, 0.965, 0.975, 0.982, 0.986]
1/16	[0, 3, 6, 9, 12, 15]	[29.89, 31.47, 32.80, 33.85, 34.58, 35.05]	[0.930, 0.952, 0.968, 0.977, 0.983, 0.987]

SwinJSCC Kodak Rayleigh

CBR	SNR	PSNR	MS-SSIM
1/96	[0, 3, 6, 9, 12, 15]	[23.37, 24.14, 24.69, 25.05, 25.25, 25.37]	[0.725, 0.778, 0.814, 0.837, 0.850, 0.858]
1/48	[0, 3, 6, 9, 12, 15]	[24.82, 25.71, 26.31, 26.71, 26.95, 27.07]	[0.794, 0.842, 0.872, 0.890, 0.901, 0.907]
1/32	[0, 3, 6, 9, 12, 15]	[25.69, 26.64, 27.22, 27.63, 27.89, 28.01]	[0.831, 0.873, 0.898, 0.913, 0.922, 0.927]
1/24	[0, 3, 6, 9, 12, 15]	[26.33, 27.30, 27.98, 28.41, 28.67, 28.81]	[0.853, 0.891, 0.914, 0.928, 0.936, 0.940]
5/96	[0, 3, 6, 9, 12, 15]	[26.80, 27.80, 28.50, 28.94, 29.21, 29.35]	[0.867, 0.904, 0.925, 0.937, 0.944, 0.948]
1/16	[0, 3, 6, 9, 12, 15]	[27.08, 28.00, 28.64, 29.03, 29.26, 29.38]	[0.878, 0.910, 0.928, 0.939, 0.945, 0.949]

HAMJSCC-Base CamVid AWGN

CBR	SNR	PSNR	MS-SSIM
1/96	[0, 3, 6, 9, 12, 15]	[25.14, 26.23, 27.02, 27.56, 27.91, 28.11]	[0.837, 0.881, 0.910, 0.929, 0.940, 0.947]
1/48	[0, 3, 6, 9, 12, 15]	[27.05, 28.38, 29.40, 30.13, 30.60, 30.89]	[0.885, 0.921, 0.944, 0.959, 0.967, 0.972]
1/32	[0, 3, 6, 9, 12, 15]	[28.28, 29.78, 30.99, 31.88, 32.49, 32.88]	[0.907, 0.938, 0.957, 0.970, 0.977, 0.981]
1/24	[0, 3, 6, 9, 12, 15]	[29.15, 30.78, 32.12, 33.16, 33.90, 34.38]	[0.920, 0.947, 0.964, 0.975, 0.981, 0.985]
5/96	[0, 3, 6, 9, 12, 15]	[29.81, 31.51, 32.93, 34.07, 34.91, 35.46]	[0.929, 0.953, 0.969, 0.978, 0.984, 0.988]
1/16	[0, 3, 6, 9, 12, 15]	[30.25, 31.89, 33.26, 34.32, 35.08, 35.57]	[0.936, 0.958, 0.972, 0.980, 0.985, 0.988]

HAMJSCC-Base Kodak Rayleigh

CBR	SNR	PSNR	MS-SSIM
1/96	[0, 3, 6, 9, 12, 15]	[23.53, 24.38, 25.03, 25.43, 25.69, 25.84]	[0.730, 0.784, 0.822, 0.848, 0.863, 0.872],
1/48	[0, 3, 6, 9, 12, 15]	[24.89, 25.86, 26.56, 27.03, 27.30, 27.40],	[0.794, 0.844, 0.877, 0.897, 0.909, 0.915]
1/32	[0, 3, 6, 9, 12, 15]	[25.76, 26.77, 27.50, 27.97, 28.26, 28.42]	[0.830, 0.873, 0.901, 0.918, 0.927, 0.933]
1/24	[0, 3, 6, 9, 12, 15]	[26.35, 27.42, 28.17, 28.64, 28.95, 29.12]	[0.850, 0.891, 0.916, 0.930, 0.939, 0.943]
5/96	[0, 3, 6, 9, 12, 15]	[26.81, 27.87, 28.65, 29.15, 29.44, 29.61]	[0.863, 0.902, 0.925, 0.938, 0.945, 0.949]
1/16	[0, 3, 6, 9, 12, 15]	[27.22, 28.22, 28.93, 29.40, 29.67, 29.81]	[0.879, 0.912, 0.932, 0.943, 0.950, 0.953]

Complete Numerical Results for Fig. 7

CamVid AWGN CBR=1/48

Scheme	PSNR	MS-SSIM
w/o-M/D (Fix SNR=8)	[20.40, 26.69, 29.09, 29.79, 30.09, 30.21]	[0.651, 0.853, 0.936, 0.950, 0.953, 0.954]
w/o-M/D (Rand SNR)	[26.24, 27.60, 28.62, 29.31, 29.71, 29.93]	[0.856, 0.901, 0.929, 0.945, 0.955, 0.960]
SNR M/D	[26.60, 27.99, 29.06, 29.80, 30.26, 30.54]	[0.871, 0.911, 0.936, 0.951, 0.961, 0.967]
SNR & CBR M/D	[27.05, 28.38, 29.40, 30.13, 30.60, 30.89]	[0.885, 0.921, 0.944, 0.959, 0.967, 0.972]

Kodak Rayleigh SNR=9

Scheme	PSNR	MS-SSIM
w/o-M/D (Fix CBR=1/32)	[15.19, 23.48, 31.26, 24.23, 20.86, 12.14]	[0.419, 0.829, 0.966, 0.845, 0.756, 0.519]
w/o-M/D (Rand CBR)	[26.61, 29.01, 30.60, 31.90, 32.81, 33.34]	[0.916, 0.948, 0.961, 0.967, 0.971, 0.974]

CBR M/D	[27.41, 30.01, 31.74, 32.98, 33.84, 34.07]	[0.928, 0.957, 0.968, 0.974, 0.977, 0.979]
SNR & CBR M/D	[27.56, 30.13, 31.88, 33.16, 34.07, 34.32]	[0.929, 0.959, 0.970, 0.975, 0.978, 0.980]

Complete Numerical Results for Fig. 8

CamVid AWGN CBR=1/48

Scheme	PSNR	LPIPS
DeepJSCC	[24.76, 25.53, 26.02, 26.70]	[0.479, 0.443, 0.425, 0.388]
ADJSCC	[24.85, 25.80, 26.48, 27.15]	[0.475, 0.444, 0.417, 0.389]
DeepJSCC-l++	[26.44, 27.33, 28.11, 28.76]	[0.404, 0.378, 0.354, 0.332]
SwinJSCC	[26.80, 27.66, 28.38, 28.98]	[0.395, 0.367, 0.342, 0.317]
HAMJSCC-Base	[27.05, 27.95, 28.75, 29.40]	[0.397, 0.367, 0.335, 0.307]
DeepJSCC-Diff	[25.06, 25.31, 25.51, 25.69]	[0.304, 0.275, 0.254, 0.238]
HAMJSCC	[26.78, 27.59, 28.26, 28.79]	[0.306, 0.281, 0.258, 0.242]

Kodak Rayleigh CBR=1/48

Scheme	PSNR	LPIPS
DeepJSCC	[23.64, 24.28, 24.70, 25.05]	[0.572, 0.535, 0.516, 0.494]
ADJSCC	[23.97, 24.55, 25.01, 25.32]	[0.559, 0.522, 0.507, 0.486]
DeepJSCC-l++	[24.47, 25.13, 25.66, 26.09]	[0.495, 0.470, 0.449, 0.432]
SwinJSCC	[24.84, 25.45, 25.96, 26.32]	[0.484, 0.464, 0.444, 0.427]
HAMJSCC-Base	[24.90, 25.56, 26.12, 26.58]	[0.491, 0.467, 0.433, 0.415]
DeepJSCC-Diff	[22.76, 23.11, 23.40, 23.61]	[0.431, 0.410, 0.397, 0.393]
HAMJSCC	[24.91, 25.43, 26.01, 26.39]	[0.414, 0.393, 0.360, 0.343]

Complete Numerical Results for Fig. 9

DeepJSCC-l++ CamVid AWGN

CBR	PSNR	LPIPS
1/96	[24.21, 25.42, 26.28]	[0.470, 0.435, 0.405]
1/48	[26.43, 27.73, 28.76]	[0.404, 0.366, 0.332]
1/32	[27.73, 29.15, 30.33]	[0.367, 0.328, 0.291]

SwinJSCC CamVid AWGN

CBR	PSNR	LPIPS
1/96	[24.78, 25.80, 26.49]	[0.454, 0.421, 0.394]
1/48	[26.80, 28.04, 28.98]	[0.394, 0.354, 0.317]
1/32	[27.99, 29.39, 30.49]	[0.364, 0.317, 0.274]

HAMJSCC-Base CamVid AWGN

CBR	PSNR	LPIPS
1/96	[25.14, 26.23, 27.02]	[0.456, 0.419, 0.387]
1/48	[27.05, 28.38, 29.40]	[0.397, 0.349, 0.307]
1/32	[28.28, 29.78, 30.99]	[0.359, 0.305, 0.260]

HAMJSCC CamVid AWGN

CBR	PSNR	LPIPS
1/96	[25.08, 26.11, 26.85]	[0.349, 0.316, 0.289]
1/48	[26.89, 28.13, 29.04]	[0.302, 0.264, 0.234]
1/32	[28.01, 29.33, 30.32]	[0.275, 0.236, 0.207]

HAMJSCC-Base Kodak Rayleigh

CBR	PSNR	LPIPS
1/96	[23.53, 24.38, 25.03, 25.43, 25.69, 25.84]	[0.558, 0.518, 0.490, 0.468, 0.451, 0.444]
1/48	[24.89, 25.86, 26.56, 27.03, 27.30, 27.40]	[0.492, 0.448, 0.415, 0.393, 0.377, 0.371]
1/32	[25.76, 26.77, 27.50, 27.97, 28.26, 28.42]	[0.448, 0.403, 0.371, 0.349, 0.335, 0.327]
1/24	[26.35, 27.42, 28.17, 28.64, 28.95, 29.12]	[0.419, 0.374, 0.342, 0.320, 0.304, 0.296]
5/96	[26.81, 27.87, 28.65, 29.15, 29.44, 29.61]	[0.399, 0.354, 0.323, 0.300, 0.284, 0.274]
1/16	[27.22, 28.22, 28.93, 29.40, 29.67, 29.81]	[0.386, 0.345, 0.314, 0.291, 0.276, 0.268]

HAMJSCC Kodak Rayleigh

CBR	PSNR	LPIPS
1/96	[23.64, 24.42, 25.01, 25.43, 25.69, 25.84]	[0.468, 0.430, 0.403, 0.468, 0.451, 0.444]
1/48	[24.91, 25.76, 26.38, 27.03, 27.30, 27.40]	[0.413, 0.372, 0.344, 0.393, 0.377, 0.371]
1/32	[25.67, 26.52, 27.12, 27.97, 28.26, 28.42]	[0.380, 0.340, 0.311, 0.349, 0.335, 0.327]
1/24	[26.35, 27.42, 28.17, 28.64, 28.95, 29.12]	[0.419, 0.374, 0.342, 0.320, 0.304, 0.296]
5/96	[26.81, 27.87, 28.65, 29.15, 29.44, 29.61]	[0.399, 0.354, 0.323, 0.300, 0.284, 0.274]
1/16	[27.22, 28.22, 28.93, 29.40, 29.67, 29.81]	[0.386, 0.345, 0.314, 0.291, 0.276, 0.268]

Complete Numerical Results for Fig. 11 and Fig. 12

HAMJSCC CamVid AWGN-AWGN

Uplink_SNR	CBR	Downlink_SNR	PSNR	LPIPS
0	1/96	[10, 20, 30]	[25.03, 25.08, 25.09]	[0.353, 0.349, 0.349]
3	1/96	[10, 20, 30]	[26.03, 26.11, 26.13]	[0.319, 0.316, 0.315]
6	1/96	[10, 20, 30]	[26.75, 26.85, 26.86]	[0.293, 0.289, 0.289]
0	1/48	[10, 20, 30]	[26.78, 26.89, 26.91]	[0.306, 0.302, 0.302]
3	1/48	[10, 20, 30]	[27.95, 28.13, 28.15]	[0.269, 0.264, 0.263]
6	1/48	[10, 20, 30]	[28.80, 29.04, 29.05]	[0.240, 0.234, 0.233]
0	1/32	[10, 20, 30]	[27.83, 28.01, 28.02]	[0.280, 0.275, 0.275]
3	1/32	[10, 20, 30]	[29.06, 29.33, 29.36]	[0.243, 0.236, 0.236]
6	1/32	[10, 20, 30]	[29.94, 30.32, 30.35]	[0.216, 0.207, 0.206]

HAMJSCC Kodak Rayleigh-AWGN

Uplink_SNR	CBR	Downlink_SNR	PSNR	LPIPS
0	1/96	[10, 20, 30]	[23.55, 23.64, 23.58]	[0.477, 0.475, 0.476]
3	1/96	[10, 20, 30]	[24.37, 24.40, 24.39]	[0.443, 0.439, 0.439]
6	1/96	[10, 20, 30]	[24.86, 24.97, 24.97]	[0.415, 0.411, 0.412]
0	1/48	[10, 20, 30]	[24.80, 24.87, 24.89]	[0.423, 0.421, 0.421]
3	1/48	[10, 20, 30]	[25.67, 25.77, 25.77]	[0.383, 0.379, 0.380]
6	1/48	[10, 20, 30]	[26.25, 26.38, 26.38]	[0.354, 0.350, 0.350]
0	1/32	[10, 20, 30]	[25.55, 25.64, 25.68]	[0.388, 0.388, 0.385]
3	1/32	[10, 20, 30]	[26.40, 26.53, 26.55]	[0.351, 0.345, 0.345]
6	1/32	[10, 20, 30]	[26.98, 27.15, 27.17]	[0.322, 0.316, 0.315]

HAMJSCC CamVid AWGN-Rician

Uplink_SNR	CBR	Downlink_SNR	PSNR	LPIPS
[0,3,6]	1/96	10	[24.93, 25.88, 26.53]	[0.349, 0.318, 0.292]
[0,3,6]	1/96	20	[24.97, 25.93, 26.61]	[0.347, 0.315, 0.290]
[0,3,6]	1/96	30	[24.98, 25.94, 26.62]	[0.347, 0.315, 0.289]
[0,3,6]	1/48	10	[26.53, 27.56, 28.26]	[0.305, 0.270, 0.244]
[0,3,6]	1/48	20	[26.61, 27.68, 28.42]	[0.302, 0.267, 0.239]
[0,3,6]	1/48	30	[26.63, 27.69, 28.44]	[0.302, 0.267, 0.239]
[0,3,6]	1/32	10	[27.42, 28.43, 29.10]	[0.281, 0.247, 0.223]
[0,3,6]	1/32	20	[27.54, 28.60, 29.32]	[0.278, 0.242, 0.217]
[0,3,6]	1/32	30	[27.55, 28.61, 29.34]	[0.277, 0.242, 0.216]

HAMJSCC Kodak Rayleigh-Rician

Uplink_SNR	CBR	Downlink_SNR	PSNR	LPIPS
[0,3,6]	1/96	10	**[23.63, 24.35, 24.90]**	**[0.468, 0.431, 0.406]**
[0,3,6]	1/96	20	**[23.65, 24.42, 24.94]**	**[0.465, 0.430, 0.404]**
[0,3,6]	1/96	30	**[23.66, 24.40, 24.95]**	**[0.465, 0.430, 0.404]**
[0,3,6]	1/48	10	**[24.80, 25.59, 26.12]**	**[0.412, 0.375, 0.348]**
[0,3,6]	1/48	20	**[24.85, 25.67, 26.24]**	**[0.412, 0.373, 0.346]**
[0,3,6]	1/48	30	**[24.89, 25.68, 26.20]**	**[0.411, 0.373, 0.345]**
[0,3,6]	1/32	10	**[25.47, 26.22, 26.74]**	**[0.383, 0.345, 0.319]**
[0,3,6]	1/32	20	**[25.56, 26.35, 26.90]**	**[0.380, 0.343, 0.315]**
[0,3,6]	1/32	30	**[25.56, 26.37, 26.91]**	**[0.380, 0.341, 0.315]**